Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Physical vector spaces for quantum mechanics

Files.

- File: `briefs/QuantumVectorSpaces/QuantumVectorSpaces.lean`
- Library: `QuantumVectorSpaces`
- Imports: `SchwartzNorms` (the L^2 seminorm family on $\mathcal{S}(\mathbb{R})$) and two new `ForMathlib` files: `ForMathlib/Analysis/InnerProductSpace/Defs.lean` (the seminorm $\sqrt{\mathrm{re}\langle x, x\rangle}$ of a `PreInnerProductSpace.Core`, and continuity of the inner product against it), `ForMathlib/Analysis/CStarAlgebra/Matrix.lean` (`Matrix.toEuclideanLin` is multiplicative)

Setting. V is a complex vector space with an inner product given as a `PreInnerProductSpace.Core`. Unlike Mathlib's `InnerProductSpace`, this puts no norm or topology on V. The defining observables are a family X_i ($i \in \iota$) in `Module.End`, and `algebra` is the unital algebra they generate (`Algebra.adjoin`). The seminorms $p_A(v) = \sqrt{\langle Av, Av\rangle}$ form `seminormFamily`, and `topology` is the topology they generate. A topology is `IsAdmissible` if it makes V a topological vector space

and makes the X_i and the inner product (jointly) continuous. For Example 3, $Xf = xf$ and $Pf = -i\hbar f'$ act on Mathlib's $\mathcal{S}(\mathbb{R}, \mathbb{C})$ (`posOp`, `momOp`), with the L^2 inner product `l2Core`. For Example 4, $S_i = (\hbar/2)\sigma_i$ acts on $\mathbb{C}^2$ as `EuclideanSpace` (`spin`).

Results.

Brief	Lean
Corollary 2, coarsest topology	`isGreatest_topology` (from `isAdmissible_topology` and `IsAdmissible.continuous_algebra`)
Corollary 2, locally convex	`locallyConvexSpace`
finite dimension (Example 4)	`topology_eq_moduleTopology`, `topology_qubit`
Example 3, $[X, P] = i\hbar$	`posOp_mul_momOp_sub`
Example 3, X, P, I symmetric	`isSymmetric_obs`
Example 3, Remark: the topology is that of $\mathcal{S}$	`topology_schwartz`, `withSeminorms_schwartz`, `isAdmissible_schwartz` (for $\hbar \neq 0$)
Example 4, $[S_i, S_j] = i\hbar\,\epsilon_{ijk} S_k$	`spin_comm`
Example 4, S_i symmetric	`isSymmetric_spin`
Theorem 5, [B]COBS	`IsAdmissible.continuous_expectation`

Differences from the brief.

- "Coarsest" is `IsGreatest` in Mathlib's order on topologies, where coarser is larger.
- The inner product is only assumed positive semidefinite in Corollary 2; definiteness is used only in finite dimension, to make the topology Hausdorff.
- Example 3 is formalized for $\mathcal{S}(\mathbb{R})$ only, not $\mathcal{S}(\mathbb{R}^n)$. The proof that the topologies agree has two halves. (a) Each seminorm $\|x^a \partial^b f\|_2$ equals $|\hbar|^{-b} p_{X^a P^b}(f)$. (b) Every element of the algebra is a combination of the normally ordered monomials $X^a P^b$ (`algebra_le_monoSpan`, from $PX^{n+1} = X^{n+1}P - (n+1)i\hbar X^n$), so every p_A is bounded by finitely many $\|x^a \partial^b f\|_2$. With `withSeminorms_l1_l2` from the previous brief, this gives the Schwartz topology.
- In finite dimension the topology of Corollary 2 is the unique Hausdorff vector-space topology, whatever the defining observables (`topology_eq_moduleTopology`). This is how Example 4 is checked.
- Not formalized: completeness and second countability in Definition 1 (Mathlib has no completeness of $\mathcal{S}$ at this pin); the "iff $n = m$" part of Example 3, which needs a notion of isomorphism of quantum vector spaces; [B]WDQ and [B]STDIST in Theorem 5 and the remarks of section 5, which are informal.

Problems found in the brief. No errors. Two steps are left implicit.

- The Remark after Example 3 says the L^2 seminorms and the sup seminorms dominate each other. But the seminorms of Corollary 2 are $\|Af\|$ for every A in the algebra, including non-normally-ordered products such as PX. Reducing them to $\|x^a \partial^b f\|$ needs the commutation relation (normal ordering), which the brief does not mention.
- Example 3 needs $\hbar \neq 0$: for $\hbar = 0$ we get $P = 0$, and the topology is just that of L^2.